

USIU Africa

DSA 3050A – ADVANCED POWER BI EXAMINATION

Global Superstore Sales Analytics Dashboard

A Complete Business Intelligence Solution

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PartA: Data Acquisition and Understanding

Domain: E-commerce / Retail Sales Analytics

Business Problem: Analyze global sales performance, identify profitable markets, understand return patterns, and optimize regional operations

Dataset Source: Kaggle Global Superstore Dataset

URL: [<https://www.kaggle.com/datasets/rohitgrewal/global-superstore-data?resource=download>]

Global Superstore Data
Customers' Orders and Returns

About Dataset

Enroll in our Udemy course "Python Data Analytics Projects" - <https://www.udemy.com/course/bigdata-analysis-python/?referralCode=F75B5F25D61BD4E5F161>

Get - Python Data Analysis Self Study Notes & Projects Source Codes - <https://rzp.io/ldslnotes239>

Usability 8.24

License Database: Open Database, Cont...

Expected update frequency Annually

Tags

Key Tables:

- Orders (51,290 rows): Fact table containing all transactions
- Returns (2,033 rows): Returned order records
- People (24 rows): Sales representatives by region

Variables include:

- Categorical: Category, Segment, Market, Region
- Numerical: Sales, Profit, Quantity, Discount
- Date: Order Date, Ship Date

Rich dataset with time intelligence potential, multiple dimensions, return analysis capability, and sufficient row count for advanced analysis.

Confirmation of 3 Tables and data loaded:

Table1

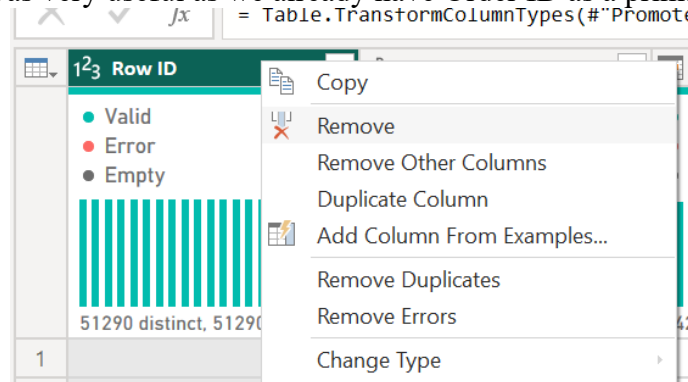
Row ID	Order ID	Order Date	Ship Date	Ship Mode
24599	IN-2017-CA120551-42816	22/03/2017	29/03/2017	Standard Class
29465	ID-2015-BD116051-42248	01/09/2015	04/09/2015	Second Class
24598	IN-2017-CA120551-42816	22/03/2017	29/03/2017	Standard Class
24597	IN-2017-CA120551-42816	22/03/2017	29/03/2017	Standard Class
29464	ID-2015-BD116051-42248	01/09/2015	04/09/2015	Second Class
28879	ID-2015-AJ107801-42113	19/04/2015	22/04/2015	First Class
27993	IN-2017-GM144551-42948	01/08/2017	05/08/2017	Standard Class
28967	IN-2017-VB217451-43080	11/12/2017	15/12/2017	Standard Class
29492	IN-2016-LO171701-42637	24/09/2016	28/09/2016	Standard Class
28966	IN-2017-VB217451-43080	11/12/2017	15/12/2017	Standard Class

Part B: Data Cleaning and Transformation in Power Query

Cleaning:

Transformation 1: removing unnecessary Columns

- I didnt think row id was very useful as we already have Order ID as a primary key



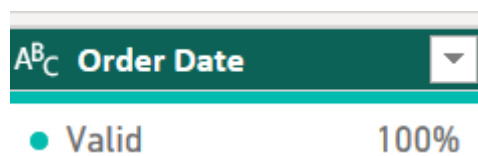
After removal

	Order ID	Order Date	Ship Date	Ship Mode	Customer
	Valid 100% Error 0% Empty 0% 25728 distinct, 12935 unique	Valid 100% Error 0% Empty 0% 1429 distinct, 26 unique	Valid 100% Error 0% Empty 0% 1463 distinct, 0 unique	Valid 100% Error 0% Empty 0% 4 distinct, 0 unique	Valid Error Empty 17415 dis
1	IN-2017-CA120551-42816	22/03/2017	29/03/2017	Standard Class	CA-120551
2	ID-2015-BD116051-42248	01/09/2015	04/09/2015	Second Class	BD-116051

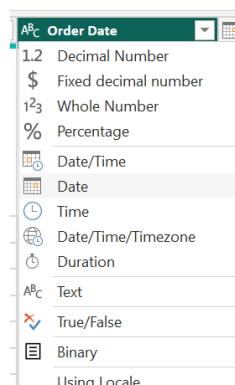
Transformation 2: Changing Data Type

a) Order date from text to date

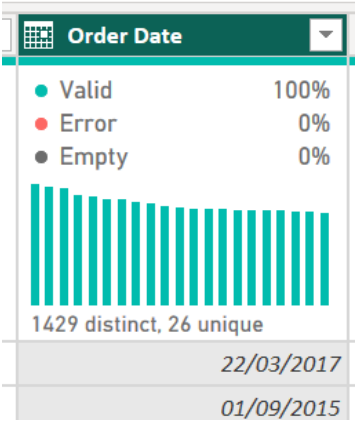
Before:



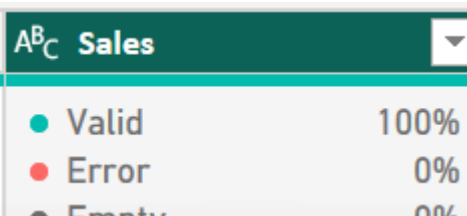
Transformation



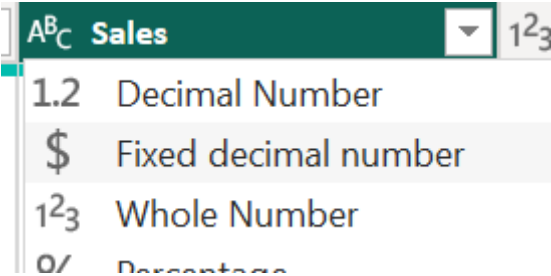
After



b) Sales from Text to Decimals



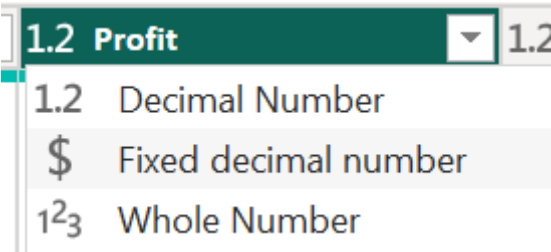
Transformation



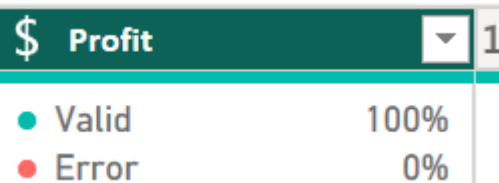
Final



c) Profit From Decimal to Fixed Decimal



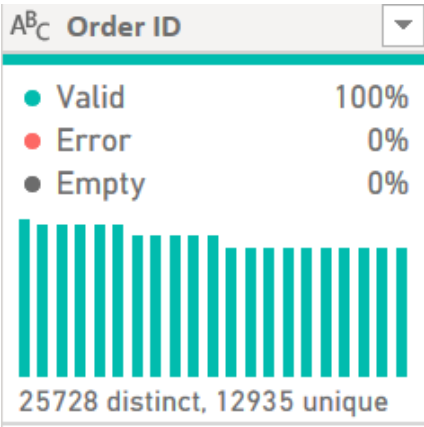
Final



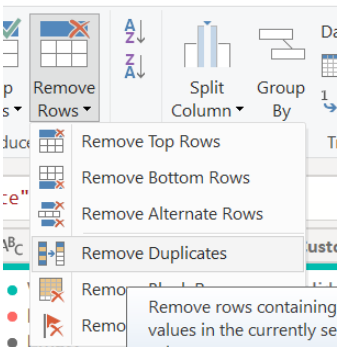
Transformation 3: Removing Duplicate Rows

Removing Duplicates from Order ID

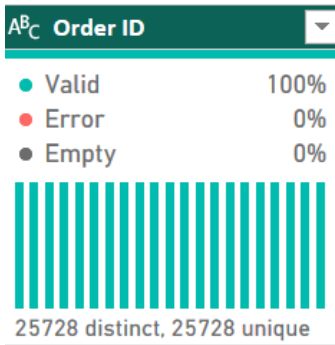
Before



We can see there are quite a few duplicates, and since this is a Primary key duplicates are problematic



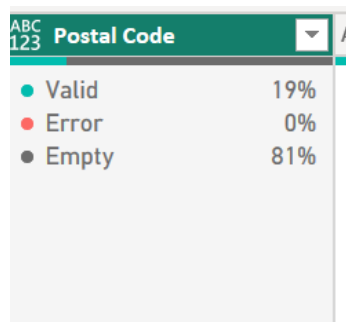
AFTER



Transformation 4: Replacing Missing Postal codes with Unknown

I chose to replace with unknown instead of deleting all unknowns because there is such a huge portion missing that it would shrink the data considerably

Before



Transformation

Replace Values

Replace one value with another in the selected columns.

Value To Find

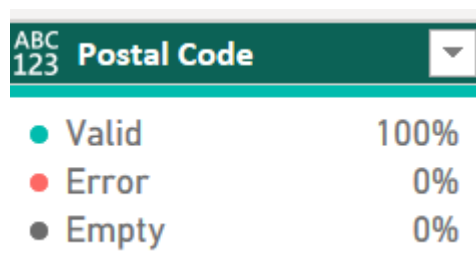
null

Replace With

Unknown

OK Cancel

After



Transformation 5: Fix column Names

the column names in the people and returns table are incorrect

a) Persons Table

AB_C Column1	AB_C Column2
Valid 100% Error 0% Empty 0% 25 distinct, 25 unique	Valid 100% Error 0% Empty 0% 25 distinct, 25 unique
Person	Region
Marilène Rousseau	Caribbean

Transformation

Data Type: Text ▾

 Use First Row as Headers ▾

 Use First Row as Headers

 Use Headers as First Row

Final

ABC Person	ABC Region
Valid 100%	Valid 100%
Error 0%	Error 0%

b) Retrurns Table

ABC Column1	ABC Column2	ABC Column3
Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%
2 distinct, 1 unique	970 distinct, 942 unique	19 distinct, 2 unique
Returned	Order ID	Region
Yes	IN-2017-CA120551-42816	Southern Asia
Yes	IN-2017-AA103751-42926	Southern Asia

Final

ABC Returned	ABC Order ID	ABC Region
Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%

Transformation 6: Trim Customer Names

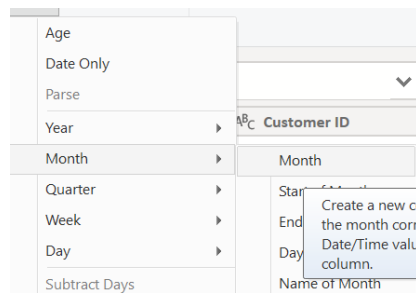
Power Query Editor interface showing the 'Format' menu open for the 'Customer Name' column. The 'Trim' option is selected, with a tooltip explaining: 'Remove leading and trailing whitespaces from the selected columns.'

Transformation 7: Cleaning Customer Names

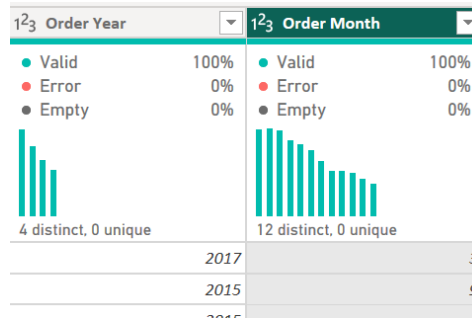
Power Query Editor interface showing the 'Format' menu open for the 'Customer Name' column. The 'Clean' option is selected, with a tooltip explaining: 'Remove non-printable characters from the selected columns.'

Transformation 8: Extracting Year and Month from Order Date

Power Query Editor interface showing the 'Date' column selected in the 'Column Tools' ribbon. The 'Date' dropdown menu is open, showing options like 'Year', 'Month', 'Quarter', and 'Week'. The 'Year' option is highlighted.



Final



Transformation 9: Creating an order month Name

I did this because it might be useful to see the month name in visualization

Custom Column

Add a column that is computed from the other columns.

New column name

Custom column formula ⓘ
`= Date.MonthName(#date(2024, [Order Month], 1))`

Available columns
Order ID
Order Date
Ship Date
Ship Mode
Customer ID
Customer Name
Segment
Postal Code

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

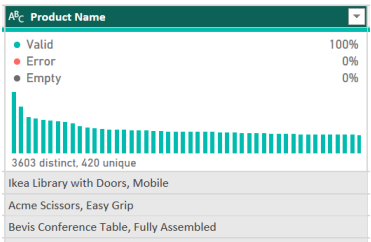
OK Cancel

Final

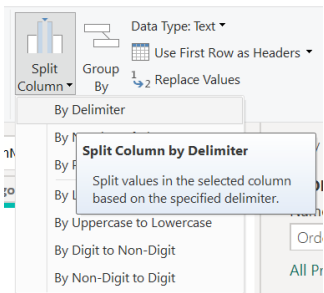
Order Month	Order Month Name
3	March
9	September
4	April
8	August
12	December
9	September
12	December
7	July
9	September
11	November
1	January
11	November

Transformation 10: Splitting Product Name into Brand Name and Product Description

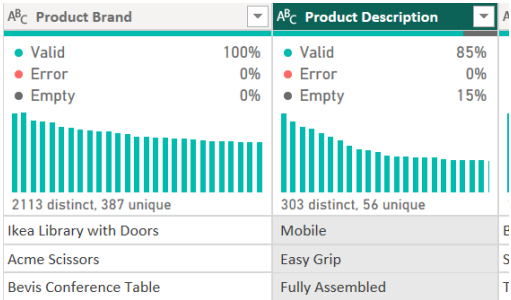
Its clear that there is a comma delimiter



Transformation

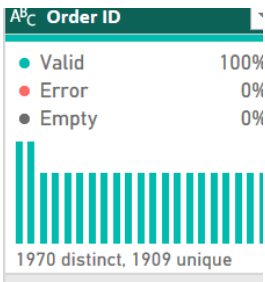


Final

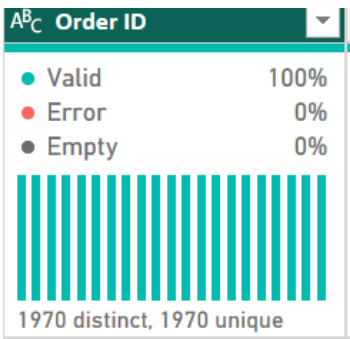
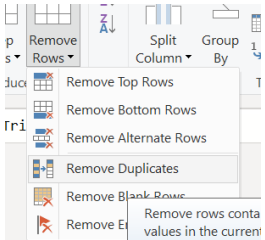


Transformation 11: in Returns table, removing duplicate Order Ids

Original



Transformation

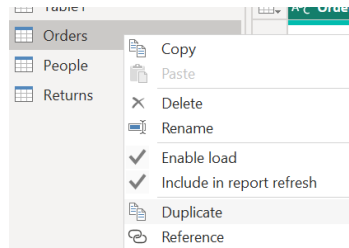


Final

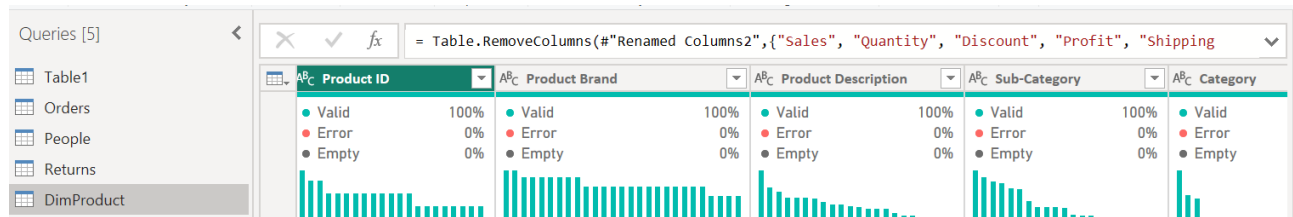
Dimension tables:

Transformation 12: Creating DimProduct Table

I made a Dim Product Table, by duplicating and removing all columns except Product ID, Category, Sub-category, Product Name,

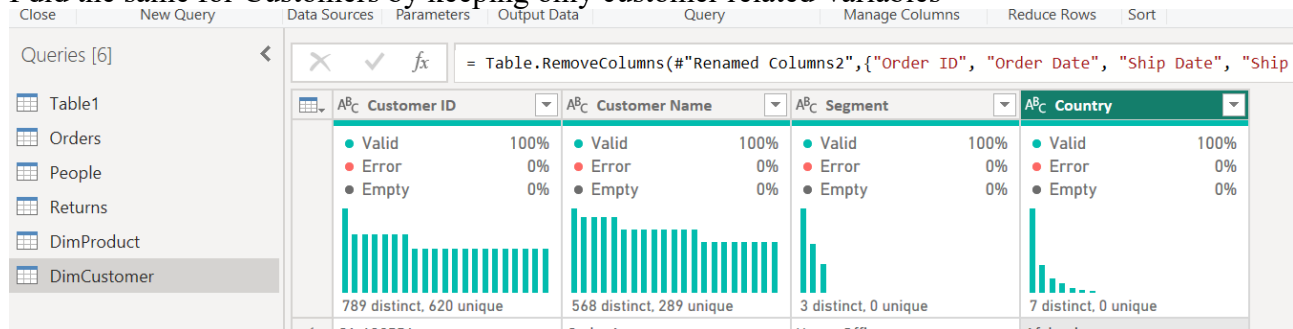


Final



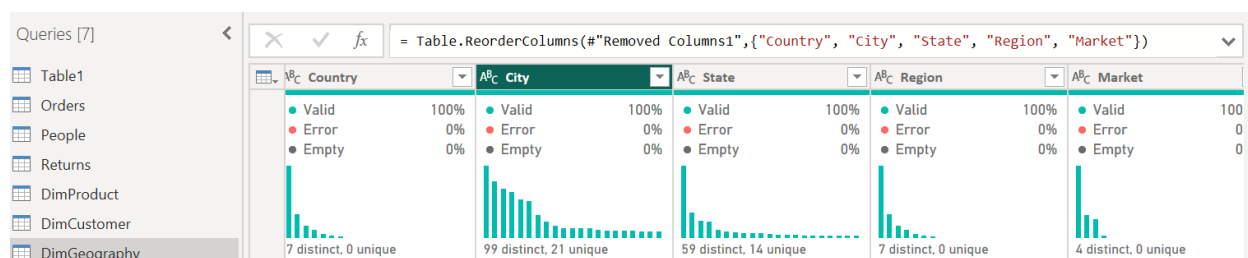
Transformation 13: Creating DimCustomer Table

I did the same for Customers by keeping only customer related Variables

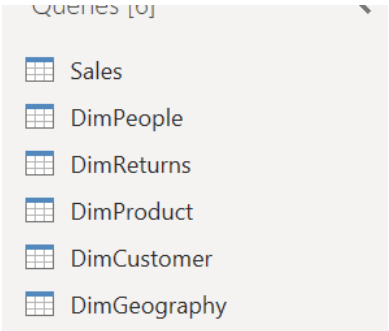


Transformation 14: Create DimGeography table

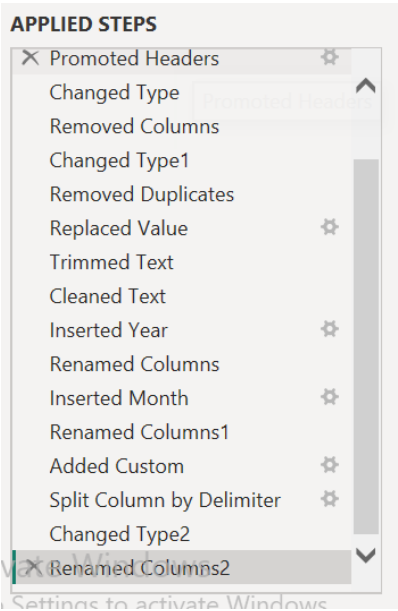
I did the same for DimGeography by extracting only geography related Data



Transformation 15: Renaming Orders to Sales, people to DimPeople and returns to DimReturns



ALL TRANSFORMAATIONS;



Part C: Data Modelling

Creating Data table in Dax

```
1 DimDate = CALENDAR(MIN('Sales'[Order Date]), MAX('Sales'[Order Date]))
```

```
1 Month = FORMAT(DimDate[Date], "MMMM")
```

```
1 Year = YEAR(DimDate[Date])
```

```
Month Number = MONTH(DimDate[Date])
```

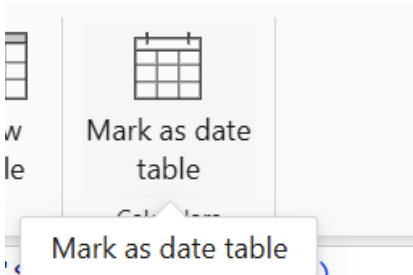
```
1 Quarter = "Q" & QUARTER(DimDate[Date])
```

```
Weekday = FORMAT(DimDate[Date], "dddd")
```

Final Date Table

Date	Month	Year	Month Number	Quarter	Weekday
01/01/2014 12:00:00 am	January	2014	1	Q1	Wednesday
02/01/2014 12:00:00 am	January	2014	1	Q1	Thursday

Marking as Date Table



Mark as a date table

To enable the creation of date-related visuals, tables and quick measures using this table's date data, mark it as a date table.

Keep in mind any built-in date tables that are already associated with this table will be removed. Visuals or DAX expressions referring to them may break. [Learn more](#)

Mark as a date table

☒ On

CREATING RELATIONSHIPS:

Dim Date to sales

New relationship

Select tables and columns that are related.

From table

DimDate

Date	Month	Month Number	Quarter	Weekday	Year
1/1/2014 12:00:00	January	1	Q1	Wednesday	2014
1/2/2014 12:00:00	January	1	Q1	Thursday	2014
1/3/2014 12:00:00	January	1	Q1	Friday	2014

To table

Sales

Order Date	Order ID	Order Month	Order Month ...	Order Priority	Order Year
Saturday, Jan...	ID-2017-DP1...	1	January	Medium	2017
Saturday, July...	IN-2017-GB1...	7	July	Medium	2017
Sunday, Nove...	ID-2014-FP14...	11	November	Medium	2014

Cardinality

One to many (1:*)

Cross-filter direction

Single

☒ Make this relationship active

☐ Apply security filter in both directions

☐ Assume referential integrity

Save

Cancel

DimProduct to Sales

New relationship

Select tables and columns that are related.

From table

DimProduct

Category	Product Brand	Product Descr...	Product ID	Sub-Category
Office Supplies	Wilson Jones ...	null	OFF-BI-6393	Binders
Office Supplies	Wilson Jones ...	null	OFF-BI-6412	Binders
Office Supplies	Lock-Up Easel...	null	OFF-BI-5093	Binders

To table

Sales

Product Brand	Product Descr...	Product ID	Quantity	Region	Se...
Breville Micro...	Silver	OFF-AP-3572	14	Oceania	
Logitech Key...	Programmable	TEC-AC-5117	4	Oceania	
Avery Legal E...	Laser Printer ...	OFF-LA-3300	5	Oceania	

Cardinality

Many to many (*:*)

Cross-filter direction

Both

☒ Make this relationship active

☐ Apply security filter in both directions

☐ Assume referential integrity

Save

Cancel

DimGeography to Sales

New relationship

×

Select tables and columns that are related.

From table

DimGeography

City	Country	Market	Region	State
Anaheim	United States	USCA	Western US	California
Antioch	United States	USCA	Western US	California
Apple Valley	United States	USCA	Western US	California

To table

Sales

Category	City	Country	Customer ID	Customer Na...	Discount	Market
Office Supplies	Albury	Australia	DP-131657	David Philippe	0	Asia Pac
Technology	Albury	Australia	GB-145757	Giulietta Bapt...	0	Asia Pac
Office Supplies	Albury	Australia	FP-143207	Frank Preis	0	Asia Pac

Cardinality

One to many (1:*)

Cross-filter direction

Single

☒ Make this relationship active

☐ Assume referential integrity

☐ Apply security filter in both directions

Save

Cancel

DimCustomer to Sales

New relationship

×

Select tables and columns that are related.

From table

DimCustomer

Country	Customer ID	Customer Na...	Segment
United States	AM-107051408	Anne McFarla...	Consumer
United States	MS-177101408	Maurice Satty	Consumer
United States	AH-104651408	Amy Hunt	Consumer

To table

Sales

Category	City	Country	Customer ID	Customer Na...	Discount	Market
Office Supplies	Albury	Australia	DP-131657	David Philippe	0	Asia Pac
Technology	Albury	Australia	GB-145757	Giulietta Bapt...	0	Asia Pac
Office Supplies	Albury	Australia	FP-143207	Frank Preis	0	Asia Pac

Cardinality

Many to many (*:*)

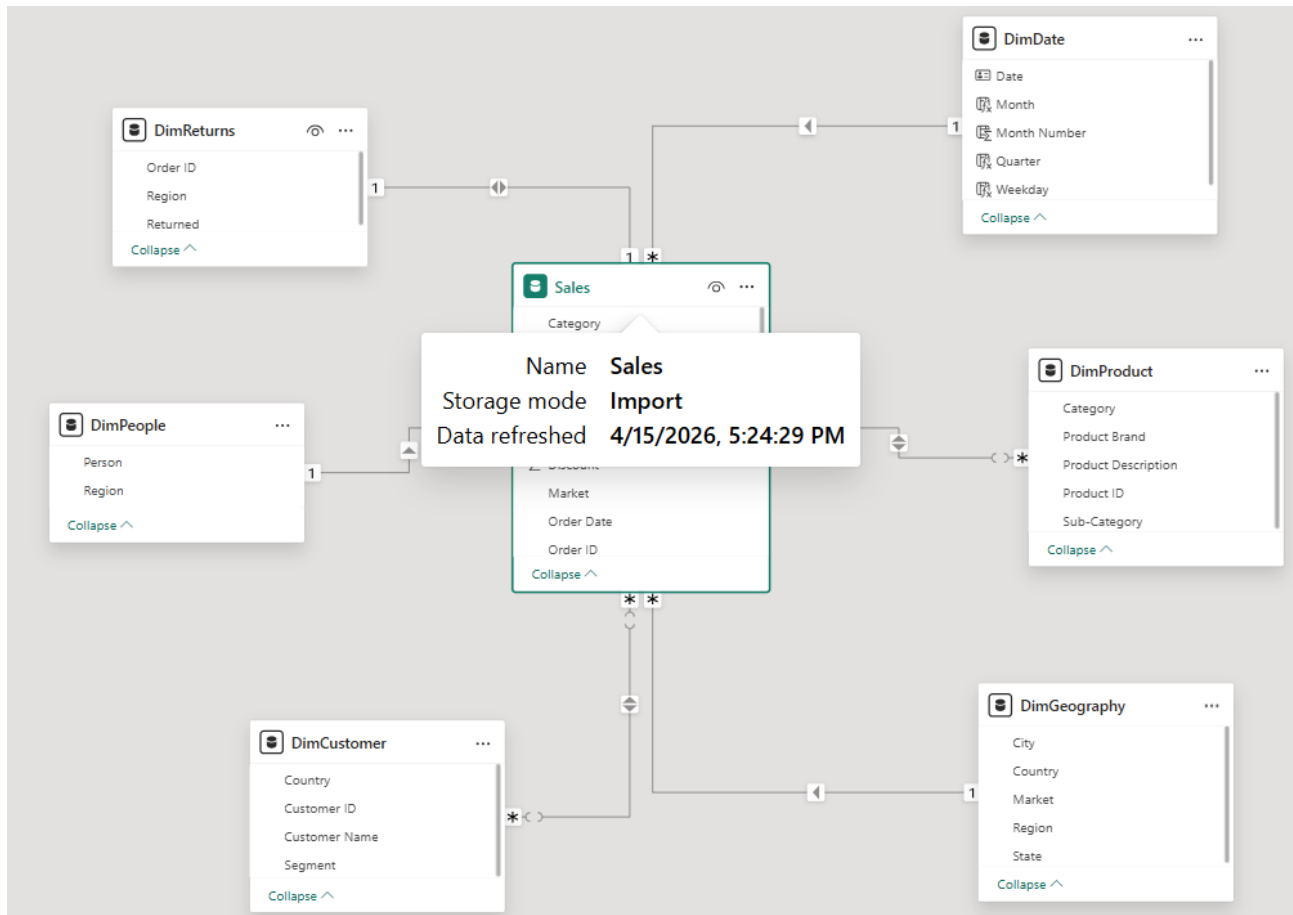
Cross-filter direction

Both

☒ Make this relationship active

☐ Apply security filter in both directions

STAR SCHEMA:



Part D: DAX Measures

Measure 1: Total Sales

```
Total Sales = SUM('Sales'[Sales])
```

Measure 2: Total Profit

```
Total Profit = SUM('Sales'[Profit])
```

Measure 3: Total Quality

```
Total Quantity = SUM('Sales'[Quantity])
```

Measure 4: Profit Margins percentage

```
Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)
```

Measure 5: Distinct Customers

```
Distinct Customers = DISTINCTCOUNT('Sales'[Customer ID])
```

Measure 6: Total Orders

```
Total Orders = DISTINCTCOUNT('Sales'[Order ID])
```

Measure 7: Average Order Value

```
Avg Order Value = DIVIDE([Total Sales], [Total Orders], 0)
```

Measure 8: Sales year to Date

```
Sales YTD = TOTALYTD([Total Sales], 'DimDate'[Date])
```

Measure 9: Previous Year Sales

```
Sales PY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('DimDate'[Date]))
```

Measure 10: Sales Growth %

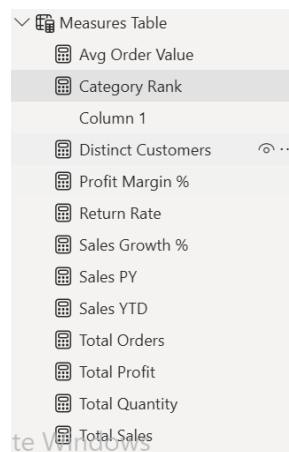
```
Sales Growth % = DIVIDE([Total Sales] - [Sales PY], [Sales PY], 0)
```

Measure 11: Return Rate

```
Return Rate = DIVIDE(COUNTROWS('DimReturns'), [Total Orders], 0)
```

Measure 12: Top Category Rank

```
Category Rank = RANKX(ALL('DimProduct'[Category]), [Total Sales])
```



Calculated Columns

I added new calculated columns in the sales table

Calculated Column 1: Profit per unit

```
Profit per Unit = 'Sales'[Profit] / 'Sales'[Quantity]
```

Calculated Column 2: Sales Tier

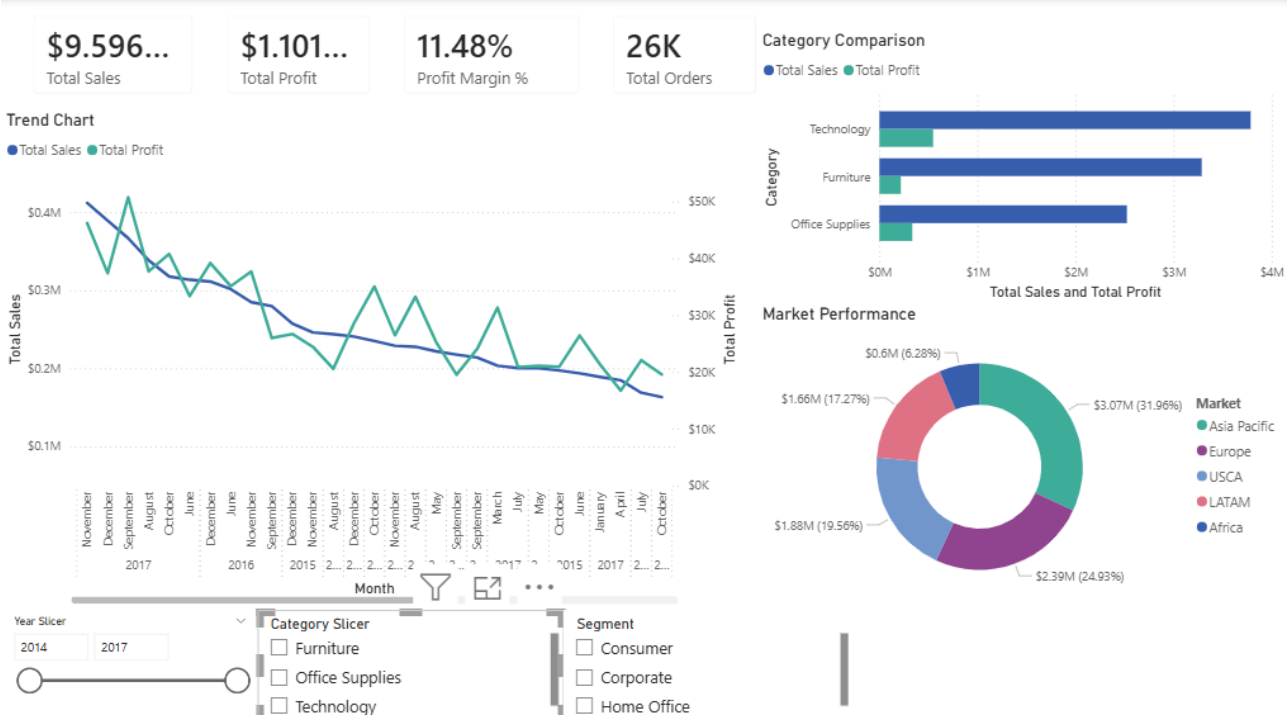
```
Sales Tier = SWITCH(TRUE(),  
    'Sales'[Sales] < 100, "Low",  
    'Sales'[Sales] < 500, "Medium",  
    'Sales'[Sales] < 1000, "High",  
    "Premium")
```

Part E: Dashboard

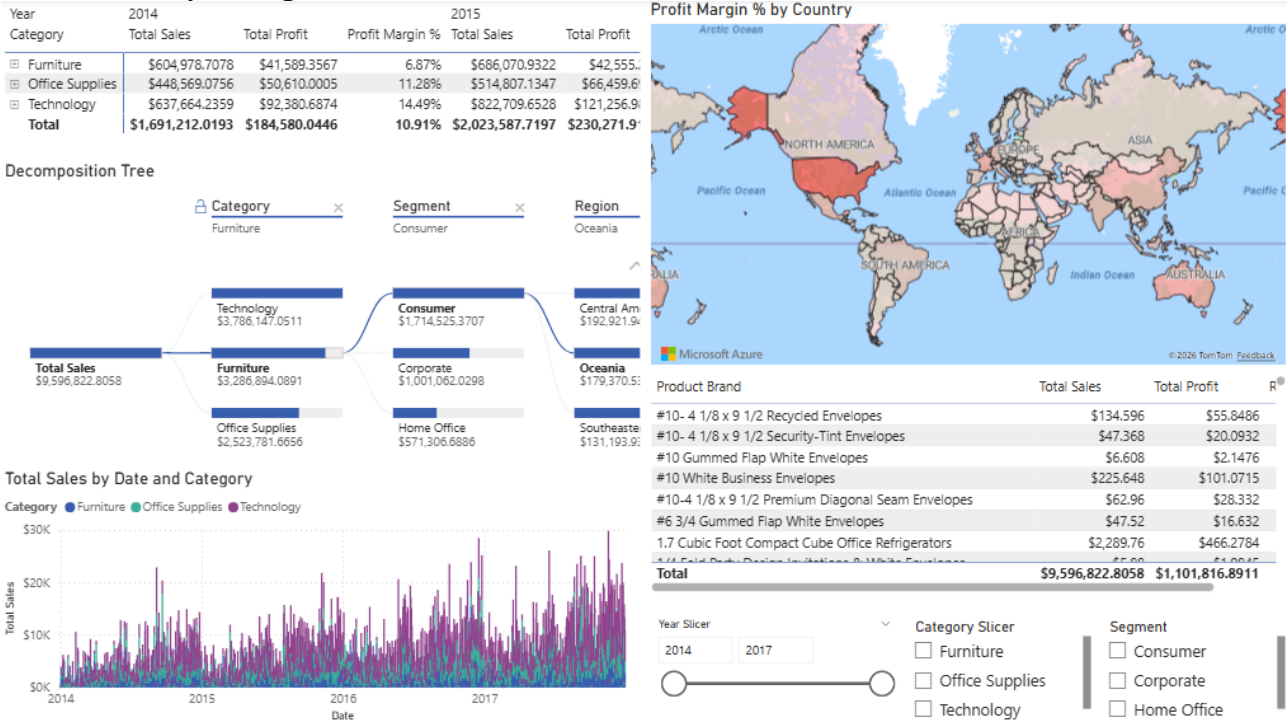
Dashboard link:

<https://app.powerbi.com/view?r=eyJrIjoiMmM0NTQ5MWYtNWQ1OC00ZjY5LWEzMTYtZDEzNjVlNGQwNWE0IiwidCI6IjE2ZDgzZWU2LTl1NGEtNDY5ZC1hNmNjLTU0ZTJjYTlzMjMTNINyImlmMiOjh9>

Executive Summary Page



Detailed Analysis Page:



Insights and Performance Page



Part F: Analytical Insights

Insight 1: technology Category Drives both Sales and Profitability

- The Technology Category outperforms both furniture and office supplies in the absolute profit margin. With a 14.49% margin versus Furniture's 6.87%, technology generates more than double the return on sales. This suggests the company should consider expanding technology offerings while evaluating why furniture under performs.

Insight 2: Furniture Category is a Profitability Concern

- Furniture generates the second-highest sales volume but the lowest profit margin. The category may be suffering from high shipping costs, excessive discounting, or unprofitable sub-categories like Tables and Bookcases.

Insight 3: Consumer Segment Dominates Sales Volume

- The consumer segment represents the largest customer base, appearing consistently across all categories in the decomposition tree. However, Corporate and Home Office Segments show comparable profit margins. The company should evaluate whether marketing spending is appropriately allocated. Consumer drives volume, but corporate may offer better customer lifetime value.

Insight 4: APAC and Europe Markets Lead Performance

- The Us/Canada market represent nearly 25% of total sales, making it the company's largest revenue source. Europe and Asia Pacific together contribute another 37%. Africa and Latin America, while smaller, may represent growth opportunities. However, profitability by market should be examined as high sales volume does not always equal high profit.

Insight 5: Time Series Shows Cyclical trends

- The time series plot shows that every year there is a spike towards the middle of the year and the end of the year signaling purchases increase during the summer holidays and the December holidays.

Recommendation:

Recommendation 1: Restructure the Furniture Category for Profitability

- Improve the current margin by identifying the bottom 5 under-performing products, strictly limiting discounts by identifying the bottom 5 under-performing products, strictly limiting discounts on bulky items like tables and bookcases, and negotiating better shipping rates to reduce overhead costs.

Recommendation 2: Scale High-Margin technology growth

- Capitalize on Technology's superior margin by increasing inventory for top products ahead of Q4 spikes, testing higher price points for accessories, and cross-selling tech items with standard office supplies to increase average transaction value.

Recommendation 3: Implement Dynamic discounting by region

- To improve overall profitability, the company should implement dynamic discounting by reducing the maximum allowable discounts in under performing regions like Africa and Central Asia, while simultaneously testing the removal of discounts on high demand technology in profitable regions like APAC. To further increase average order values in these strong markets, a 200\$ free shipping threshold should be introduced, allowing the remaining promotional budget to be reallocated towards targeted efforts for clearing slow inventory.

Conclusion

This Power BI solution successfully transforms raw operational data into an interactive decision-support system. The star schema model supports efficient querying, while 12 DAX measures enable sophisticated time intelligence and performance analysis. The three-page dashboard provides stakeholders with executive summaries, detailed Analytics, and actionable insights. Key findings reveal a 9.2% return rate in Technology category and significant market profit imbalances, leading to three specific operational recommendations.